

Model ↓ / OOD →	DetectRL (Wu et al., 2024)				MultiSocial (Macko et al., 2025)				RAID (Dugan et al., 2024)				TSM (Quaremba et al., 2026)			
	ArXiv	Reddit	Yelp	News	en	de	ru	zh	Cohere	GPT4	Llama	Mistral	FP	PE	SUM	TST
TextFluoroscopy	0.7580	0.9054	0.8691	<u>0.9402</u>	0.8818	0.8423	0.8437	<u>0.8212</u>	0.8402	0.9296	0.9236	0.9478	0.4052	0.5629	0.4807	0.4747
BiScope	0.8641	0.8426	0.9289	0.8545	0.7568	0.6565	0.6995	0.5169	0.9039	0.8482	0.9594	0.9294	0.6923	0.5744	0.6867	0.6692
RepreGuard	0.5327	0.5967	0.7687	0.5187	0.5047	0.5460	0.5020	0.5020	0.7047	0.7553	0.7367	0.7673	0.5007	0.5227	0.5793	0.4873
RoBERTa	0.9108	0.8735	0.8491	0.7781	0.8511	0.7870	0.7784	0.6389	0.9118	0.9554	0.9688	0.9503	0.8324	0.7367	0.8452	0.7376
LLP	0.9830	0.9800	0.9897	0.9441	0.9321	0.8820	0.8945	0.8696	0.9267	0.9812	0.9908	0.9821	0.9513	0.8087	0.9512	0.8560
Δ vs BL	+7.22	+7.46	+6.08	+0.39	+5.03	+3.96	+5.08	+4.84	+1.49	+2.58	+2.20	+3.17	+11.89	+7.20	+10.60	+11.83
CLP	<u>0.9807</u>	<u>0.9665</u>	<u>0.9867</u>	0.9263	<u>0.9087</u>	<u>0.8651</u>	<u>0.8759</u>	0.8043	<u>0.9200</u>	<u>0.9723</u>	<u>0.9839</u>	<u>0.9791</u>	<u>0.9317</u>	<u>0.7943</u>	<u>0.9152</u>	<u>0.8451</u>
Δ vs BL	+6.99	+6.11	+5.78	-1.40	+2.69	+2.27	+3.22	-1.69	+0.82	+1.69	+1.51	+2.88	+9.93	+5.76	+7.01	+10.75

Table 2: OOD detection mean AUC scores across four benchmarks, comparing the four strongest detectors from Table 1 against MGT probes. OOD columns report the average transfer performance across the three remaining subsets within each benchmark (e.g., the ArXiv column averages Reddit → ArXiv, Yelp → ArXiv, and News → ArXiv). **Bold** indicates the best score per column, underline the second-best score, and “Δ vs. BL” the gain over the strongest baseline. Appendix Figure 10 presents the full OOD results. **Takeaway:** MGT probes exhibit strong OOD transferability, outperforming training-based detectors by +0.39–11.37 AUC on average.

5.3 Out-of-domain (OOD) Detection

If probes recover genuine MGT signals rather than dataset artifacts—a key limitation of existing supervised detectors (Doughman et al., 2025)—they should transfer OOD. **RQ2:** *How well do MGT probes compare to existing detectors in OOD settings?*

MGT probes preserve strong OOD transferability, outperforming training-based detectors by 0.39–11.37 AUC. Table 2 reports mean OOD AUC for each target subset not seen during training. Each column averages transfer performance from the three remaining subsets within the same benchmark. For example, the ArXiv column averages Reddit → ArXiv, Yelp → ArXiv, and News → ArXiv.⁵ As baselines, we include the strongest training-based detectors from Table 1. Overall, MGT probes transfer effectively across benchmark settings, improving over baselines by up to 11.83 AUC. LLP consistently exhibits more robust transfer performance than CLP.

On DetectRL, LLP exhibits strong OOD transferability across domains, outperforming baselines by 0.39–7.46 AUC. CLP achieves similarly strong transfer except on News, which we attribute to overfitting to stylistic cues associated with formal writing.⁶ On MultiSocial, LLP demonstrates consistent and reliable cross-lingual transfer, even between typologically distant languages (e.g., ru → zh). We interpret this finding as evidence for a language-agnostic MGT direction encoded across layers. In contrast, CLP exhibits weaker cross-lingual transfer and fails to generalize from ru → zh, suggesting that its globally learned representation is less robust

for OOD transfer. As in the ID setting, baselines already perform strongly on RAID’s cross-generator transfer. Nevertheless, both probes achieve modest but consistent improvements.

The largest gains occur in cross-task transfer, where probes outperform baselines by 5.76–11.89 AUC. Similar to the ID setting, supervised detectors struggle to generalize across tasks, particularly on PE and TST. Despite these challenges, both probes maintain strong performance, achieving AUC > 0.80 in these difficult settings.

5.4 Probing Vector Similarity

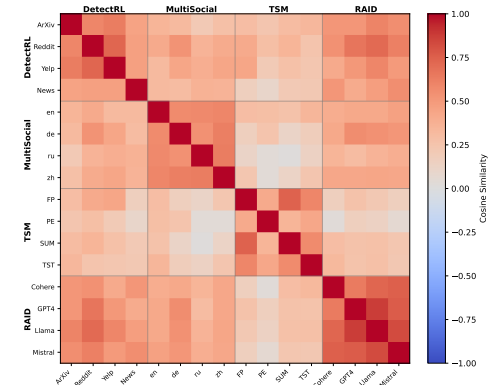


Figure 4: Cosine similarities of probing vectors at layer 16 across all 16 datasets. **Takeaway:** Probing vectors exhibit high similarity both within and across datasets, suggesting the existence of a shared latent MGT direction.

To better understand the strong transferability of probes, we ask **RQ3:** *Why do MGT probes generalize across OOD settings?* Prior work has identified *universal* probing directions that can be recovered across settings (Agarwal et al., 2025; Wang et al., 2026). We therefore analyze the alignment of MGT probing vectors at the layer with the highest OOD

⁵Appendix Figure 10 reports the full OOD transfer matrix.

⁶We further analyze CLP’s transfer behavior on News and Chinese in Appendix C.2.

AUC performance (see Appendix Figure 15).

Probing vectors exhibit high within-benchmark and moderate cross-benchmark similarity, suggesting a shared latent MGT direction. Figure 4 shows the pairwise cosine similarities of LLP probing vectors at layer 16 across all 16 evaluation subsets. We find that probing vectors are highly aligned within benchmarks and moderately aligned across benchmarks. This suggests that probes recover a relatively stable latent MGT direction, helping explain the strong OOD transfer observed in Table 2. Additionally, Appendix Figure 11 illustrates that latent human-machine directions are approximately parallel across subsets of DetectRL. Together, “machineness” appears to be encoded as a shared and therefore transferable linear direction in latent space.

5.5 Sample-efficiency Analysis

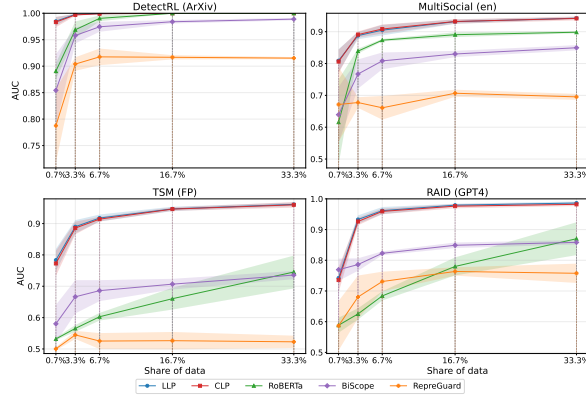


Figure 5: In-domain AUC scores for selected datasets as a function of training set size. **Takeaway:** MGT probes achieve strong performance with only 10–100 training samples.

Given the probes’ simplicity and the low-dimensional, linear MGT signal, we expect them to require fewer labeled examples than more complex detectors, hence **RQ4: How sample-efficient are MGT probes compared to existing MGT detectors?**

MGT probes achieve strong performance with only 10–100 training samples while exhibiting low sampling uncertainty. Figure 5 plots AUC as a function of training set size for one subset from each benchmark.⁷ To quantify sampling uncertainty, we repeat each experiment with different random seeds. Compared to the strongest baseline,

⁷We still fit the PCA subspace for each layer using the full training set. Appendix Figure 12 shows nearly identical results without PCA-reduced activations.

RoBERTa, both probes improve rapidly, reaching near-peak performance with only 3.3–6.7% of the training data (10–100 samples) before plateauing. Moreover, probes exhibit substantially lower sampling uncertainty than RepreGuard and RoBERTa, indicating that the latent MGT direction can be recovered reliably even in low-resource settings.

5.6 Detecting varying degrees of AI-editing

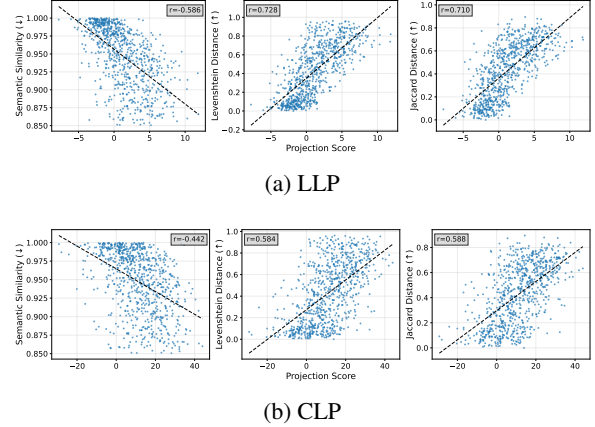


Figure 6: Edit-strength metrics for samples from APT-Eval (Saha and Feizi, 2025) plotted against probe projection scores. The dashed line shows the linear regression fit, and r denotes the Pearson correlation coefficient. **Takeaway:** Projection scores correlate strongly with edit-strength metrics, suggesting that the latent MGT direction captures a continuous spectrum of “machineness.”

While the previous experiments focus on binary detection, a growing body of work seeks to identify human-written text that has been edited by AI (Zhang et al., 2024). Probe projection scores provide a natural continuous measure of MGT, which raises **RQ5: Do MGT probes encode fine-grained signals of AI-polished text?** To test this hypothesis, we use APT-Eval (Saha and Feizi, 2025) and EditLens (Thai et al., 2026), which contain AI-edited texts with varying degrees of edits. We correlate probe projection scores with three metrics that quantify the degree of AI editing (cosine similarity, Levenshtein and Jaccard distance) as in Saha and Feizi (2025).

Probe projection scores strongly correlate with the degree of AI editing, indicating that probing vectors capture fine-grained levels of AI-polished text. Figure 6 plots the text-similarity metrics against projection scores for LLP (top) and CLP (bottom). The dashed line shows a linear regression fit, and r denotes the Pearson correlation.

tion coefficient. LLP exhibits strong correlations across all similarity metrics, indicating that human text subjected to stronger AI editing is projected closer to the machine-text region of latent space. Crucially, this behavior emerges despite the probes being trained only with binary labels. In contrast, CLP shows only moderate correlations, which we attribute to the difficulty of capturing subtle AI edits in a joint space. These findings suggest that the latent MGT direction represents a *continuous spectrum* of “machineness,” enabling fine-grained MGT detection beyond binary classification. Appendix Figure 14 shows similar results for EditLens.

6 Conclusion

In this work, we provide empirical evidence that machine-generated text (MGT) and human-written text occupy linearly separable regions in latent space, and offer a potential explanation through analyses of representation quality metrics. We further show that simple MGT probes—linear probes trained on LM hidden-state activations—achieve stronger OOD robustness while requiring fewer training samples than supervised detectors. Finally, we show that the continuous probing direction captures fine-grained levels of AI editing.

Overall, our findings suggest that “machineness” is encoded as a stable, continuous, and linearly accessible direction in activation space, enabling both robust detection and fine-grained estimation of AI-edited text. Future work could explore alternative probing variants, investigate how these representations change under adversarial attacks, and further extend the approach to AI-edited text detection.

Limitations

Non-Exhaustive Analysis of Representational Differences We analyze HWT and MGT representations using four representation-quality metrics. However, these metrics capture only selected information-theoretic and geometric properties of latent representations, and therefore provide a non-exhaustive view of their distributional differences. While the observed differences help explain the existence of a linear decision boundary, other important representational properties may also contribute to the separation between HWT and MGT. Nevertheless, our analysis offers an initial characterization of how human- and machine-written text differ in latent space.

Model Selection Although we ablate probe performance across model architectures and sizes, we do not systematically study how detection performance scales with model size or architectural design choices. Future work could investigate these factors in greater depth and analyze how they influence the emergence and transferability of MGT signals.

Linear Probing Variants We focus on the simplest probe architecture: logistic regression. Prior work has proposed more expressive alternatives, including attention-based probes and learned pooling strategies, which may recover richer MGT signals. Exploring such probe variants represents a promising direction for future research.

Universality of the Latent MGT Direction Although our results show that MGT and HWT are linearly separable and that linear probes transfer effectively across OOD settings, they do not establish the existence of a *universal* latent MGT direction. Representation spaces contain many potential confounding factors, and probing provides only one method for identifying latent concepts. Future work should further investigate the stability, causality, and universality of MGT directions across models and data distributions.

Fine-Grained Detection of AI-Edited Text While we show that probe projection scores correlate strongly with the degree of AI editing, we do not evaluate probes as dedicated fine-grained detectors or compare them against specialized multiclass or regression-based approaches. Our goal is to establish that the latent MGT direction encodes meaningful variation in AI involvement. Future work could optimize probe training for continuous prediction and benchmark their competitiveness on fine-grained AI-editing tasks.

Limited applicability to accessible models Linear probes (Alain and Bengio, 2016) require access to model internals. A limitation of MGT probes is therefore that they are only applicable to open-source models for which hidden states are accessible.

Ethical Considerations

We do not identify any risks arising from this work. We use the AI assistant ChatGPT for proofreading and for assistance with formatting tables and figures.